

MAXIMUM LIKELIHOOD ESTIMATION (MLE)

Powerful method to frame estimation as an optimization problem.

$\Rightarrow$ Let $\underline{X}_1, \dots, \underline{X}_n$ have a joint probability function:

$$f(x_1, \dots, x_n | \theta) = P(\underline{X}_1 = x_1, \dots, \underline{X}_n = x_n | \theta)$$

$\rightarrow$ If $\underline{X}_i$'s are i.i.d. then $f(x_1, x_2, \dots, x_n | \theta) = \prod_{i=1}^n f(x_i | \theta)$

$$lik(\theta) = f(x_1, x_2, \dots, x_n | \theta) = \prod_{i=1}^n f(\underline{X}_i | \theta)$$

$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} lik(\theta) \Rightarrow l(\theta) = \log[lik(\theta)] = \log \prod_{i=1}^n f(\underline{X}_i | \theta) =$
 $= \sum_{i=1}^n \log f(\underline{X}_i | \theta)$
 $= \underset{\theta}{\operatorname{argmax}} l(\theta)$
easier to optimize a sum

EXAMPLE $\underline{X}_i \sim \text{Poisson}(\lambda)$

$$f(x_1, x_2, \dots, x_n | \theta) = \prod_{i=1}^n f(x_i | \theta) = \prod_{i=1}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$$

$$\begin{aligned} l(\theta) &= \log(lik(\theta)) = \log(f(\underline{X}_1, \underline{X}_2, \dots, \underline{X}_n | \theta)) \\ &= \log \prod_{i=1}^n \frac{\lambda^{\underline{X}_i} e^{-\lambda}}{\underline{X}_i!} = \\ &= \sum_{i=1}^n \left[\underline{X}_i \log(\lambda) + (-\lambda) - \log(\underline{X}_i!) \right] \end{aligned}$$

$\rightarrow$ to maximize $l(\lambda)$

$$l'(\lambda) = \frac{1}{\lambda} \sum \underline{X}_i - n = 0$$

$$\frac{1}{\lambda} \sum \underline{X}_i = n \Rightarrow \hat{\lambda} = \frac{1}{n} \sum_{i=1}^n \underline{X}_i = \bar{\underline{X}}$$

sample mean!
Same as method of moments

EXAMPLE: $\underline{X}_i \sim \text{normal}(\mu, \sigma^2)$ *unknown $\theta = [\mu, \sigma^2]$*

$$f(x_1, x_2, \dots, x_n | \mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x_i - \mu}{\sigma} \right)^2}$$

$$\log(f) = \sum_{i=1}^n \left[-\log(\sigma) - \frac{1}{2} \log(2\pi) - \frac{1}{2} \left(\frac{x_i - \mu}{\sigma} \right)^2 \right]$$

$$l(\mu, \sigma^2) = \log(f) |_{\underline{X}_i} = -n \log(\sigma) - \frac{n}{2} \log(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n (\underline{X}_i - \mu)^2$$

$$\rightarrow \frac{\partial l}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (\underline{X}_i - \mu) = \frac{1}{\sigma^2} (n \bar{\underline{X}} - n \mu) \left\{ \begin{array}{l} \frac{\partial l}{\partial \mu} = 0 \Rightarrow \boxed{\hat{\mu} = \bar{\underline{X}}} \text{ — MLEs} \\ \frac{\partial l}{\partial \sigma} = 0 \Rightarrow \boxed{\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (\underline{X}_i - \bar{\underline{X}})^2} \end{array} \right.$$

$$\rightarrow \frac{\partial l}{\partial \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n (\underline{X}_i - \mu)^2$$

$\frac{n}{\sigma} = \frac{1}{\sigma^3} \sum_{i=1}^n (\underline{X}_i - \bar{\underline{X}})^2$